

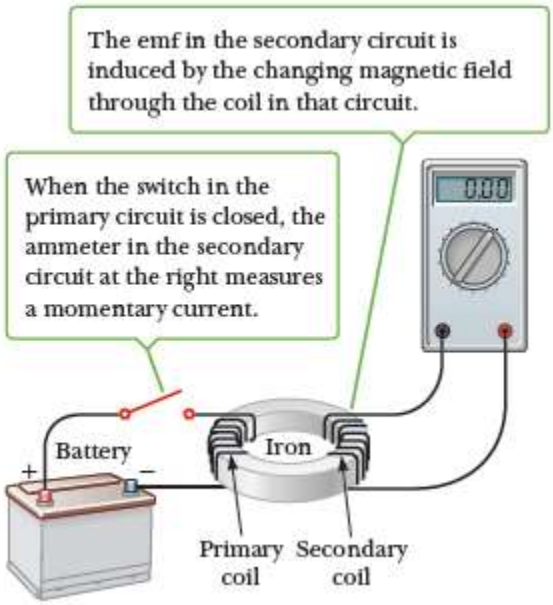
6 Induced Voltages and Inductance

In 1819 Hans Christian Oersted discovered that an electric current exerted a force on a magnetic compass. Although there had long been speculation that such a relationship existed, Oersted's finding was the first evidence of a link between electricity and magnetism. Because nature is often symmetric, the discovery that electric currents produce magnetic fields led scientists to suspect that magnetic fields could produce electric currents. Indeed, experiments conducted by Michael Faraday in England and independently by Joseph Henry in the United States in 1831 showed that a changing magnetic field could induce an electric current in a circuit. The results of these experiments led to a basic and important law known as Faraday's law. In this chapter we discuss Faraday's law and several practical applications, one of which is the production of electrical energy in power generation plants throughout the world.

6.1 Induced emf and Magnetic Flux

An experiment first conducted by Faraday demonstrated that a current can be produced by a changing magnetic field. The apparatus shown in Figure 6.1 consists of a coil connected to a switch and a battery. We call this coil the primary coil and the corresponding circuit the primary circuit. The coil is wrapped around an iron ring to intensify the magnetic field produced by the current in the coil. A second coil, called the secondary coil, at the right, is wrapped around the iron ring and is connected to an ammeter. The corresponding circuit is called the secondary circuit. It's important to notice that *there is no battery in the secondary circuit*.

At first glance, you might guess that no current would ever be detected in the secondary circuit. When the switch in the primary circuit in Figure 6.1 is suddenly closed, however, something amazing happens: the ammeter measures a current in the secondary circuit and then returns to zero! When the switch is opened again, the ammeter reads a current in the opposite direction and again returns to zero. Finally, whenever there is a steady current in the primary circuit, the ammeter reads zero.



From such observations, Faraday concluded that an electric current could be produced by a changing magnetic field. (A steady magnetic field doesn't produce a current unless the coil is moving, as explained below.) The current produced in the secondary circuit occurs for only an instant while the magnetic field through the secondary coil is changing. In effect, the secondary circuit behaves as though a source of emf were connected to it for a short time. It's customary to say that **an induced emf is produced in the secondary circuit by the changing magnetic field**.

Magnetic Flux

To evaluate induced emfs quantitatively, we need to understand what factors affect the phenomenon. Although changing magnetic fields always induce electric fields, in other situations the magnetic field remains constant, yet an induced electric field is still produced. The best example of this is an electric generator: A loop of conductor rotating in a constant magnetic field creates an electric current.

The physical quantity associated with magnetism that creates an electric field is a **changing magnetic flux**. Magnetic flux is defined in the same way as electric flux and is proportional to both the strength of the magnetic field passing through the plane of a loop of wire and the area of the loop.

The **magnetic flux** Φ_B through a loop of wire with area A is defined by

$$\Phi_B \equiv B_\perp A = BA\cos\theta \qquad [6.1]$$

where $B_\perp$ is the component of a uniform magnetic field $\vec{B}$ perpendicular to the plane of the loop, as in Figure 2.2a, and θ is the angle between $\vec{B}$ and the normal (perpendicular) to the plane of the loop.

SI unit: weber (Wb)

Note that there are always two directions normal to a given plane surface. In Figure 2.2, for example, that direction could be chosen to be to the right, resulting in positive flux. The normal direction could also be chosen to point to the left, which would result in a negative flux of the same magnitude. The choice of normal direction is called the orientation of the surface. Once chosen in a given problem, the normal direction remains fixed. A good default choice is to choose the normal direction so that the initial angle between the magnetic field and the normal direction is less than 90°.

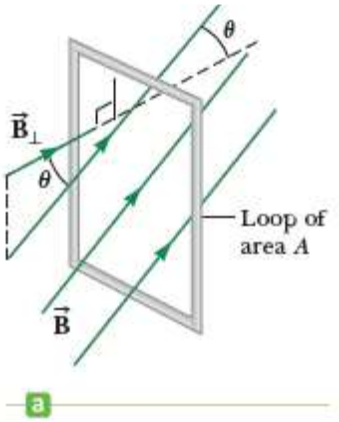
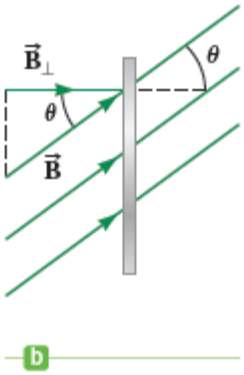


Figure 6.2



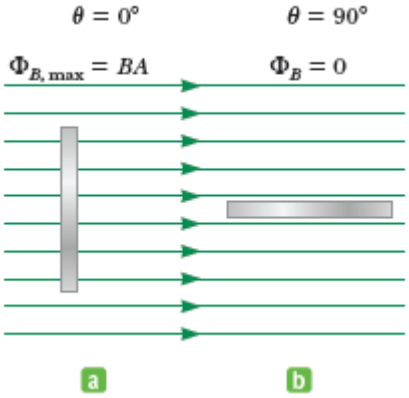


Figure 6.3

From Equation 6.1, it follows that $B_\perp = B\cos\theta$. The magnetic flux, in other words, is the magnitude of the part of $\vec{B}$ that is perpendicular to the plane of the loop times the area of the loop. Figure 6.2b is an edge view of the loop and the penetrating magnetic field lines. When the field is perpendicular to the plane of the loop as in Figure 6.3a, $\theta = 0$ and Φ_B has a maximum value, $\Phi_{Bmax} = BA$. When the plane of the loop is parallel to $\vec{B}$ as in Figure 6.3b, $\theta = 90°$ and $\Phi_B = 0$. The flux can also be negative. For example, when $\theta = 180°$, the flux is equal to -BA. Because the SI unit of B is the tesla, or weber per square meter, the unit of flux is $T.m^2$, or weber (Wb).

We can emphasize the qualitative meaning of Equation 6.1 by first drawing magnetic field lines, as in Figure 6.3. The number of lines per unit area increases as the field strength increases. The value of the magnetic flux is proportional to the total number of lines passing through the loop. We see that the most lines pass through the loop when its plane is perpendicular to the field, as in Figure 6.3a, so the flux has its maximum value at that time. As Figure 6.3b shows, no lines pass through the loop when its plane is parallel to the field, so in that case $\Phi_B = 0$.

PROBLEM A conducting circular loop of radius 0.250 m is placed in the xy-plane in a uniform magnetic field of 0.360 T that points in the positive z-direction, the same direction as the normal to the plane. (a) Calculate the magnetic flux through the loop. (b) Suppose the loop is rotated clockwise around the x-axis, so the normal direction now points at a 45.0° angle with respect to the z-axis. Recalculate the magnetic flux through the loop. (c) What is the change in flux due to the rotation of the loop?

STRATEGY After finding the area, substitute values into the equation for magnetic flux for each part. Because the normal direction was chosen to be the same direction as the magnetic field, the angle between the magnetic field and the normal is initially 0°. After the rotation, that angle becomes 45°.

SOLUTION

(a) Calculate the initial magnetic flux through the loop.

First, calculate the area of the loop:

$$A = \pi r^2 = \pi(0.250\ \text{m})^2 = 0.196\ \text{m}^2$$

Substitute A, B, and $\theta = 0°$ into Equation 20.1 to find the initial magnetic flux:

$$\Phi_B = AB\cos\theta = (0.196\ \text{m}^2)(0.360\ \text{T})\cos(0°)$$
$$= 0.070\,6\ \text{T}\cdot\text{m}^2 = \boxed{0.070\,6\ \text{Wb}}$$

(b) Calculate the magnetic flux through the loop after it has rotated 45.0° around the x-axis.

Make the same substitutions as in part (a), except the angle between $\vec{B}$ and the normal is now $\theta = 45.0°$:

$$\Phi_B = AB\cos\theta = (0.196\ \text{m}^2)(0.360\ \text{T})\cos(45.0°)$$
$$= 0.049\,9\ \text{T}\cdot\text{m}^2 = \boxed{0.049\,9\ \text{Wb}}$$

(c) Find the change in the magnetic flux due to the rotation of the loop.

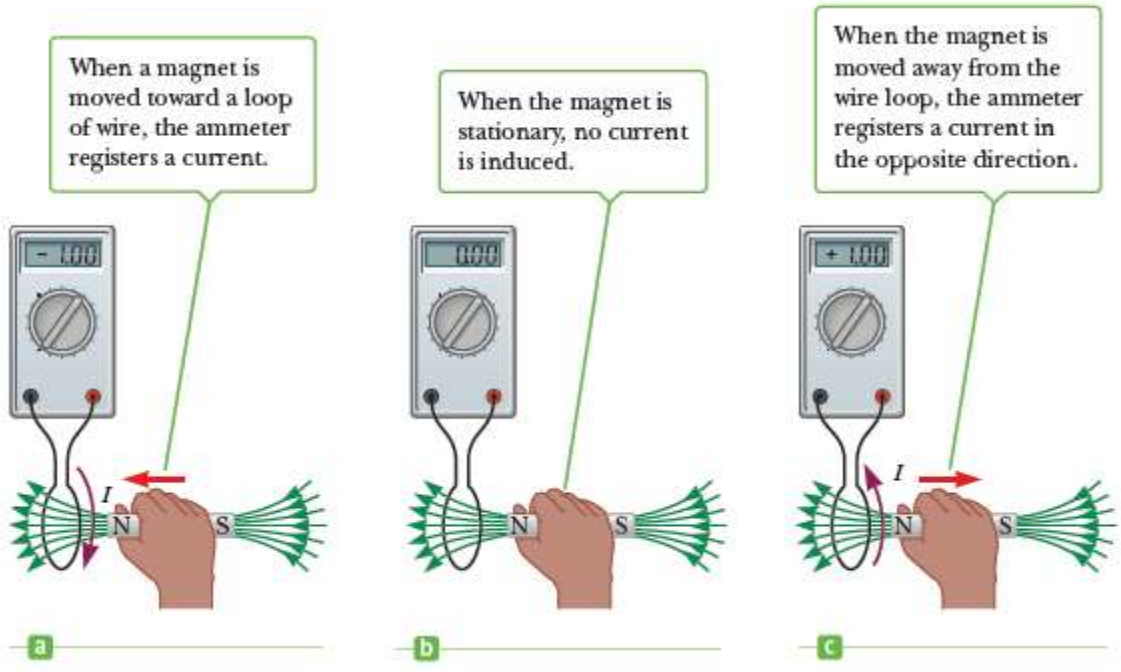
Subtract the result of part (a) from the result of part (b):

$$\Delta\Phi_B = 0.049\,9\ \text{Wb} - 0.070\,6\ \text{Wb} = \boxed{-0.020\,7\ \text{Wb}}$$

6.2 Faraday's Law of Induction and Lenz's Law

The usefulness of the concept of magnetic flux can be made obvious by another simple experiment that demonstrates the basic idea of electromagnetic induction. Consider a wire loop connected to an ammeter as in Active Figure 6.4. If a magnet is moved toward the loop, the ammeter reads a current in one direction, as in Active Figure 20.4a. When the magnet is held stationary, as in Active Figure 20.4b, the ammeter reads zero current. If the magnet is moved away from the loop, the ammeter reads a current in the opposite direction, as in Active Figure 20.4c. If the magnet is held stationary and the loop is moved either toward or away from the magnet, the ammeter also reads a current. From these observations, it can be concluded that a current is set up in the circuit as long as there is relative motion between the magnet and the loop. The same experimental results are found whether the loop moves or the magnet moves. We call such a current an induced current because it is produced by an induced emf.

This experiment is similar to the Faraday experiment discussed in Section 6.1. In each case, an emf is induced in a circuit when the magnetic flux through the circuit changes with time. It turns out that the instantaneous emf induced in a circuit equals the negative of the rate of change of magnetic flux with respect to time through the circuit. This is Faraday's law of magnetic induction.



Active Figure 6.4

If a circuit contains N tightly wound loops and the magnetic flux through each loop changes by the amount $\Delta\Phi_B$ during the interval Δt, the average emf induced in the circuit during time Δt is

$$\mathcal{E} = -N\frac{\Delta\Phi_B}{\Delta t} \qquad [6.2]$$

Because $\Phi_B = BA\cos\theta$, a change of any of the factors B, A, or θ with time produces an emf. We explore the effect of a change in each of these factors in the following sections. The minus sign in Equation 6.2

is included to indicate the polarity of the induced emf. This polarity determines which of two directions current will flow in a loop, a direction given by **Lenz's law**:

> *The current caused by the induced emf travels in the direction that creates a magnetic field with flux opposing the change in the original flux through the circuit.*

Lenz's law says that if the magnetic flux through a loop is becoming more positive, say, then the induced emf creates a current and associated magnetic field that produces negative magnetic flux. Some mistakenly think this "counter magnetic field" created by the induced current, called $\overrightarrow{B_{ind}}$ ("ind" for induced), will always point in a direction opposite the applied magnetic field $\vec{B}$, but that is only true half the time! Figure 6.5a shows a field penetrating a loop. The graph in Figure 6.5b shows that the magnitude of the magnetic field $\vec{B}$ shrinks with time, which means that the flux of $\vec{B}$ is shrinking with time, so the induced field $\overrightarrow{B_{ind}}$ will actually be in the same direction as $\vec{B}$. In effect, $\overrightarrow{B_{ind}}$ "shores up" the field $\vec{B}$, slowing the loss of flux through the loop.

The direction of the current in Figure 6.5a can be determined by right-hand rule number 2: Point your right thumb in the direction that will cause the fingers on your right hand to curl in the direction of the induced field $\overrightarrow{B_{ind}}$. In this case, that direction is counterclockwise: with the right thumb pointed in the direction of the current, your fingers curl down outside the loop and around and up through the inside of the loop. Remember, inside the loop is where it's important for the induced magnetic field to be pointing up.

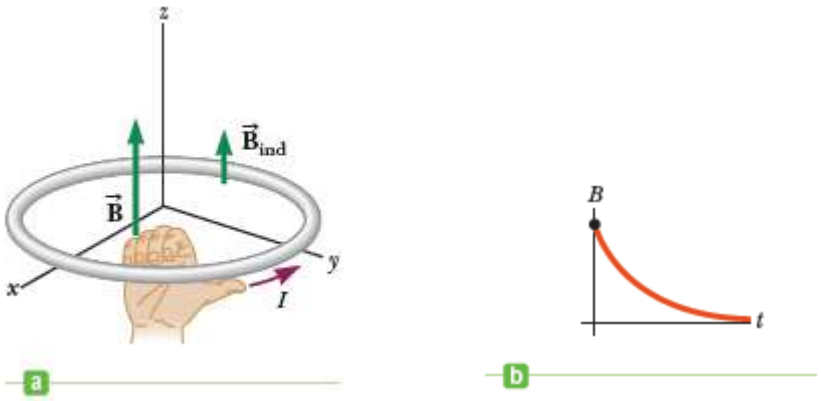


Figure 6.5

PROBLEM A coil with 25 turns of wire is wrapped on a frame with a square cross section 1.80 cm on a side. Each turn has the same area, equal to that of the frame, and the total resistance of the coil is 0.350 V. An applied uniform magnetic field is perpendicular to the plane of the coil, as in Figure 6.6. (a) If the field changes uniformly from 0.00 T to 0.500 T in 0.800 s, what is the induced emf in the coil while the field is changing? Find (b) the magnitude and (c) the direction of the induced current in the coil while the field is changing.

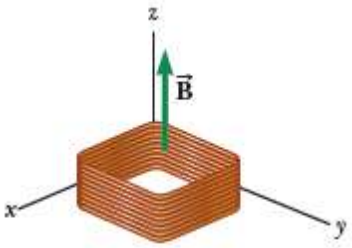


Figure 6.6

STRATEGY Part (a) requires substituting into Faraday's law. The necessary information is given, except for DFB, the change in the magnetic flux during the elapsed time. Using the normal direction to coincide with the positive z-axis, compute the initial and final magnetic fluxes with Equation 6.1, find the difference, and assemble all terms in Faraday's law. The current can then be found with Ohm's law, and its direction with Lenz's law.

SOLUTION

(a) Find the induced emf in the coil.

To compute the flux, the area of the coil is needed:

$$A = L^2 = (0.018\ 0\ \text{m})^2 = 3.24 \times 10^{-4}\ \text{m}^2$$

The magnetic flux $\Phi_{B,i}$ through the coil at $t = 0$ is zero because $B = 0$. Calculate the flux at $t = 0.800$ s:

$$\Phi_{B,f} = BA\cos\theta = (0.500\ \text{T})(3.24 \times 10^{-4}\ \text{m}^2)\cos(0°) = 1.62 \times 10^{-4}\ \text{Wb}$$

Compute the change in the magnetic flux through the cross section of the coil over the 0.800-s interval:

$$\Delta\Phi_B = \Phi_{B,f} - \Phi_{B,i} = 1.62 \times 10^{-4}\ \text{Wb}$$

Substitute into Faraday's law of induction to find the induced emf in the coil:

$$\mathcal{E} = -N\frac{\Delta\Phi_B}{\Delta t} = -(25\ \text{turns})\left(\frac{1.62 \times 10^{-4}\ \text{Wb}}{0.800\ \text{s}}\right) = -5.06 \times 10^{-3}\ \text{V}$$

(b) Find the magnitude of the induced current in the coil.

Substitute the voltage difference and the resistance into Ohm's law, where $\Delta V = \mathcal{E}$:

$$I = \frac{\Delta V}{R} = \frac{5.06 \times 10^{-3}\ \text{V}}{0.350\ \Omega} = 1.45 \times 10^{-2}\ \text{A}$$

(c) Find the direction of the induced current in the coil.

The magnetic field is increasing up through the loop, in the same direction as the normal to the plane; hence, the flux is positive and is also increasing. A downward-pointing induced magnetic field will create negative flux, opposing the change. If you point your right thumb in the clockwise direction along the loop as viewed from above, your fingers curl down through the loop, which is the correct direction for the counter magnetic field. Hence the current must proceed in a clockwise direction as viewed from above the coil.

Finding the Direction of the Induced Current

Finding the direction of the induced current can be tricky. The following three examples illustrate how the direction is found using Lenz's law.

Lenz's Law Example 1 The current in the wire of Figure 6.7 is steadily increasing in the direction indicated. Let's choose the normal direction to be out of the page so that magnetic field vectors coming out of the page will produce positive magnetic flux.

The magnetic field created by the current I circulates around the wire, going into the page on the right side of the long wire in the region of the rectangular coil and coming out of the page on the left side of the long straight wire. Therefore, the magnetic flux through the rectangular coil due to the current I is negative. Because the current is increasing up the page, the magnetic field is becoming stronger, increasing the magnitude of the negative flux through the rectangular coil. By Lenz's law, the induced current in the coil must produce positive flux, countering the increasing negative fl ux. That requires an induced magnetic field pointing out of the page through the coil. Mentally curl the fingers of the right hand around the right branch of the rectangular coil, and note that the fingers come straight up out of the page through the coil, as required. The right thumb, meanwhile, points up the page, indicating the current direction in that part of the coil. Therefore the induced current in the coil is counterclockwise.

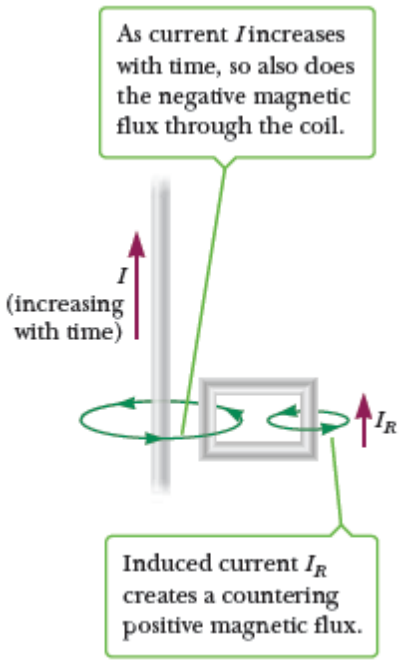


Figure 6.7

Lenz's Law Example 2 In Figure 6.8a, the north pole of the magnet moves toward the coil. If the normal direction is chosen to the right, then the magnetic flux through the coil due to the magnet is positive and increases with time.

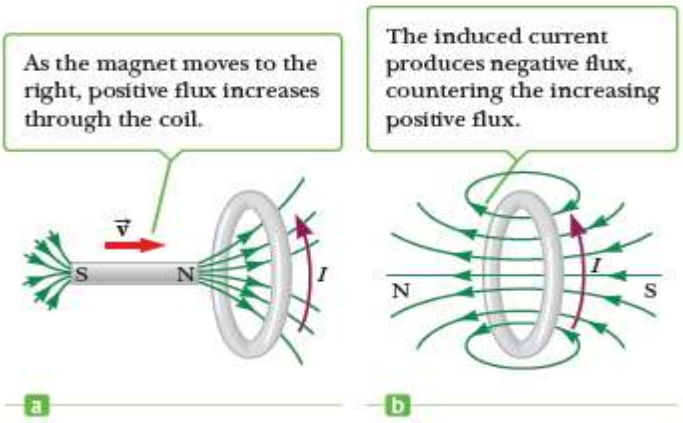


Figure 6.8

A negative flux to the left must therefore be created by the induced current in the coil, so the induced magnetic field must also point to the left as indicated in Figure 6.8b. Imagine curling the fingers of the right hand around the coil so they point through the coil to the left. The right thumb then points upward, indicating the induced current is counterclockwise as viewed from the left end of the loop.

Lenz's Law Example 3 Consider a coil of wire placed near a solenoid in Figure 6.9a. The wire is wrapped in such a way as to create a south magnetic pole at the right end when the switch is closed in Figure 6.9b. Choose the normal direction to the left. When the switch is closed, the current in the solenoid begins to increase, and the magnetic flux through the coil is positive and increasing with time. Therefore the induced current in the coil must create negative magnetic flux to counteract the increasing positive flux created by the current in the solenoid. That requires an induced magnetic field directed to the right through the coil. Turning the right hand so the thumb is pointed downward, the right fingers can curl through the coil and to the right. The induced current in the coil follows the direction of the right thumb, which is clockwise as viewed from the left end of the coil. When the switch is opened again in Figure 20.10c the current in the solenoid changes direction because the magnetic field and positive flux begin to decrease. Moving counterclockwise, the induced current creates positive flux through the coil, opposing the decrease in positive flux.

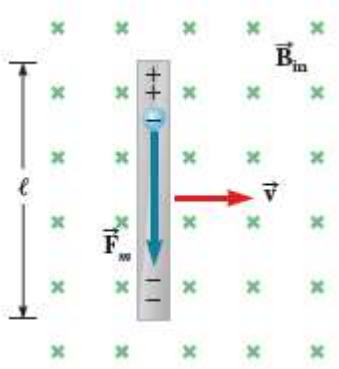
In all three of these examples, the critical idea is that a changing flux causes an induced current, and the associated induced magnetic field produces flux opposing the change in flux in accordance with Lenz's law. When the flux stops changing, the induced current stops. Although in each case the magnetic flux changed because of a changing magnetic field, induced currents can result even when the magnetic field is constant provided the flux through the loop changes. That fact will become clear when discussing motional emf in Section 6.3 and generators in Section 6.4.

6.3 Motional emf

In Section 6.2 we considered emfs induced in a circuit when the magnetic field changes with time. In this section we describe a particular application of Faraday's law in which a so-called motional emf is produced. It is the emf induced in a conductor moving through a magnetic field.

First consider a straight conductor of length l, moving with constant velocity through a uniform magnetic field directed into the paper, as in Figure 6.14. For simplicity, we assume the conductor moves in a direction perpendicular to the field. A magnetic force of magnitude $F_m = qvB$, directed downward, acts on the electrons in the conductor. Because of this magnetic force, the free electrons move to the lower end of the conductor and accumulate there, leaving a net positive charge at the upper end. As a result of this charge separation, an electric field is produced in the conductor. The

charge at the ends builds up until the downward magnetic force qvB is balanced by the upward electric force qE. At this point, charge stops flowing and the condition for equilibrium requires that



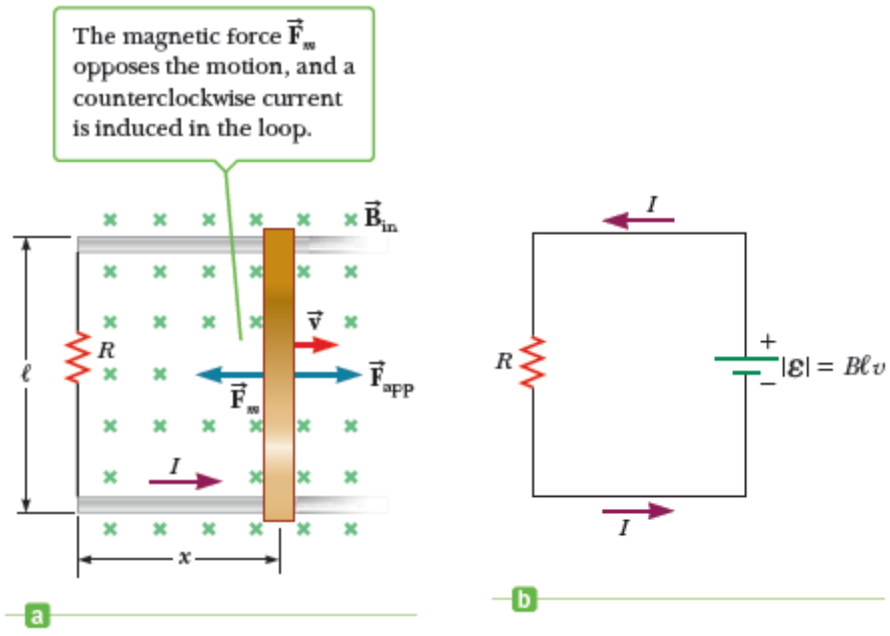
$$qE = qvB \qquad \text{or} \qquad E = vB$$

Because the electric field is uniform, the field produced in the conductor is related to the potential difference across the ends by $\Delta V = El$, giving

$$\Delta V = E\ell = B\ell v \qquad [6.3]$$

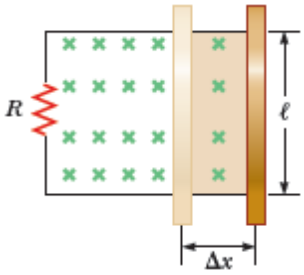
Because there is an excess of positive charge at the upper end of the conductor and an excess of negative charge at the lower end, the upper end is at a higher potential than the lower end. There is a potential difference across a conductor as long as it moves through a field. If the motion is reversed, the polarity of the potential difference is also reversed.

A more interesting situation occurs if the moving conductor is part of a closed conducting path. This situation is particularly useful for illustrating how a changing loop area induces a current in a closed circuit described by Faraday's law. Consider a circuit consisting of a conducting bar of length l, sliding along two fixed, parallel conducting rails, as in Active Figure 6.15a. For simplicity, assume the moving bar has zero resistance and the stationary part of the circuit has constant resistance R. Take the normal direction to coincide with the z-axis, out of the page. A uniform and constant magnetic field $\vec{B}$ is applied perpendicular to the plane of the circuit. As the bar is pulled to the right in the positive x-direction with velocity $\vec{V}$ under the influence of an applied force $\vec{F}_{app}$, a magnetic force along the length of the bar acts on the free charges in the bar. This force in turn sets up an induced current because the charges are free to move in a closed conducting path. In this case, the changing magnetic flux through the loop and the corresponding induced emf across the moving bar arise from the change in area of the loop as the bar moves through the magnetic field. Because the flux into the page increases, by Lenz's law the induced current circulates counterclockwise, producing flux out of the page that opposes the change.



Active Figure 6.15

Assume the bar moves a distance Δx in time Δt, as shown in Figure 6.16. The increase in flux DFB through the loop in that time is the amount of flux that now passes through the portion of the circuit that has area , Δx:



$$\Delta\Phi_B = BA = B\ell\,\Delta x$$

Using Faraday's law and noting that there is one loop (N = 1), we find that the magnitude of the induced emf is

$$|\mathcal{E}| = \frac{\Delta\Phi_B}{\Delta t} = B\ell\frac{\Delta x}{\Delta t} = B\ell v \qquad [6.4]$$

Active Figure 6.16

This induced emf is often called a **motional emf** because it arises from the motion of a conductor through a magnetic field. Further, if the resistance of the circuit is R, the magnitude of the induced current in the circuit is

$$I = \frac{|\mathcal{E}|}{R} = \frac{B\ell v}{R} \qquad [6.5]$$

Active Figure 6.15b shows the equivalent circuit diagram for this example.

PROBLEM An airplane with a wingspan of 30.0 m flies due north at a location where the downward component of Earth's magnetic field is 0.600×10^{-4} T. There is also a component pointing due north that has a magnitude of 0.470×10^{-4} T. (a) Find the difference in potential between the wingtips when the speed of the plane is 2.50×10^{2} m/s. (b) Which wingtip is positive?

STRATEGY Because the plane is flying north, the northern component of magnetic field won't have any effect on the induced emf. The induced emf across the wing is caused solely by the downward component of the Earth's magnetic field. Substitute the given quantities into Equation 6.4. Use right-hand rule number 1 to find the direction positive charges would be propelled by the magnetic force.

SOLUTION

(a) Calculate the difference in potential across the wingtips.

Write the motional emf equation and substitute the given quantities:

$$\mathcal{E} = B\ell v = (0.600 \times 10^{-4}\ \text{T})(30.0\ \text{m})(2.50 \times 10^{2}\ \text{m/s})$$
$$= 0.450\ \text{V}$$

(b) Which wingtip is positive?

Apply right-hand rule number 1:

Point the fingers of your right hand north, in the direction of the velocity, and curl them down, in the direction of the magnetic field. Your thumb points west. Therefore the west wingtip is positive.

PROBLEM (a) The sliding bar in Figure 6.15a has a length of 0.500 m and moves at 2.00 m/s in a magnetic field of magnitude 0.250 T. Using the concept of motional emf, find the induced voltage in the moving rod. (b) If the resistance in the circuit is 0.500 V, find the current in the circuit and the power delivered to the resistor. (Note: The current in this case goes counterclockwise around the loop.) (c) Calculate the magnetic force on the bar. (d) Use the concepts of work and power to calculate the applied force.

STRATEGY For part (a), substitute into Equation 6.4 for the motional emf. Once the emf is found, substitution into Ohm's law gives the current. In part (c), use Equation 5.6 for the magnetic force on a current-carrying conductor. In part (d), use the fact that the power dissipated by the resistor multiplied by the elapsed time must equal the work done by the applied force.

SOLUTION

(a) Find the induced emf with the concept of motional emf.

Substitute into Equation 20.4 to find the induced emf: $\mathcal{E} = B\ell v = (0.250\ \text{T})(0.500\ \text{m})(2.00\ \text{m/s}) = 0.250\ \text{V}$

(b) Find the induced current in the circuit and the power dissipated by the resistor.

Substitute the emf and the resistance into Ohm's law to find the induced current: $I = \dfrac{\mathcal{E}}{R} = \dfrac{0.250\ \text{V}}{0.500\ \Omega} = 0.500\ \text{A}$

Substitute $I = 0.500$ A and $\mathcal{E} = 0.250$ V into Equation 17.8 to find the power dissipated by the 0.500-Ω resistor: $P = I\,\Delta V = (0.500\ \text{A})(0.250\ \text{V}) = 0.125\ \text{W}$

(c) Calculate the magnitude and direction of the magnetic force on the bar.

Substitute values for I, B, and ℓ into Equation 19.6, with $\sin\theta = \sin(90°) = 1$, to find the magnitude of the force: $F_m = IB\ell = (0.500\ \text{A})(0.250\ \text{T})(0.500\ \text{m}) = 6.25 \times 10^{-2}\ \text{N}$

Apply right-hand rule number 2 to find the direction of the force: Point the fingers of your right hand in the direction of the positive current, then curl them in the direction of the magnetic field. Your thumb points in the negative x-direction.

(d) Find the value of F_{app}, the applied force.

Set the work done by the applied force equal to the dissipated power times the elapsed time: $W_{app} = F_{app}d = P\Delta t$

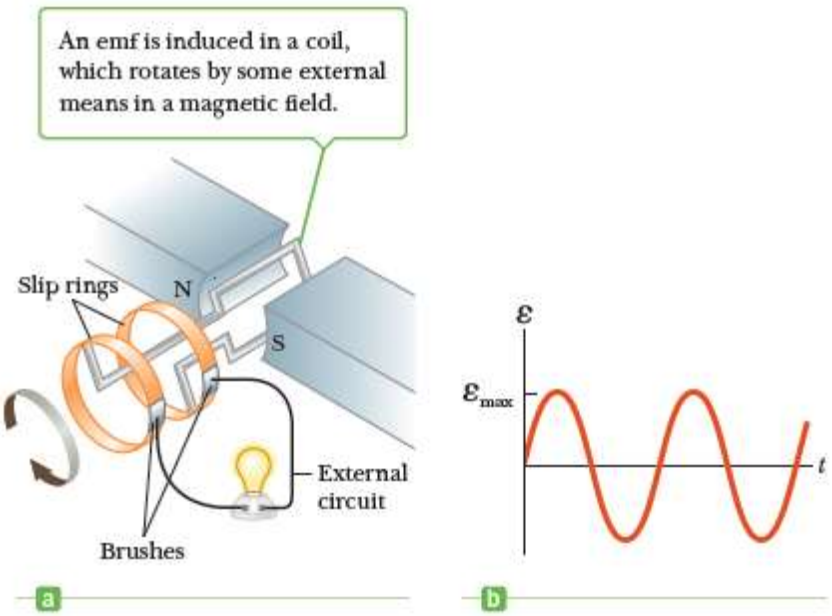
Solve for F_{app} and substitute $d = v\,\Delta t$: $F_{app} = \dfrac{P\Delta t}{d} = \dfrac{P\Delta t}{v\Delta t} = \dfrac{P}{v} = \dfrac{0.125\ \text{W}}{2.00\ \text{m/s}} = 6.25 \times 10^{-2}\ \text{N}$

REMARKS Part (d) could be solved by using Newton's second law for an object in equilibrium: Two forces act horizontally on the bar and the acceleration of the bar is zero, so the forces must be equal in magnitude and opposite in direction. Notice the agreement between the answers for F_m and F_{app}, despite the very different concepts used.

6.4 Generators

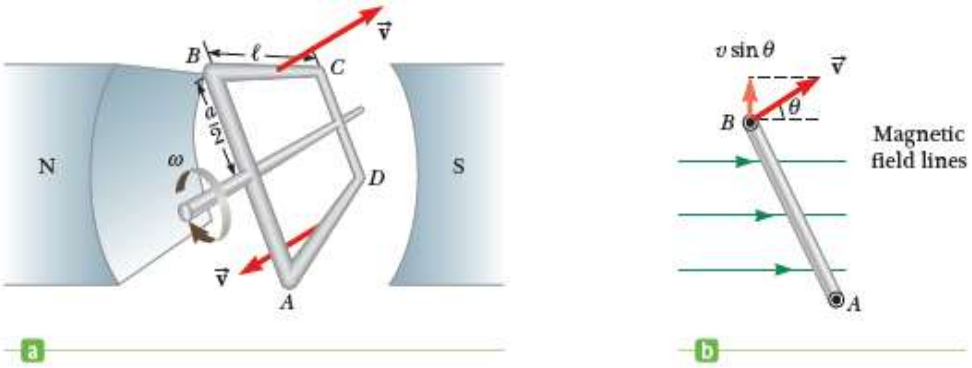
Generators and motors are important practical devices that operate on the principle of electromagnetic induction. First, consider the alternating-current (AC) generator, a device that converts mechanical energy to electrical energy. In its simplest form, the AC generator consists of a wire loop rotated in a magnetic field by some external means (Active Fig. 6.18a). In commercial power plants, the energy required to rotate the loop can be derived from a variety of sources. In a hydroelectric plant, for example, falling water directed against the blades of a turbine produces the rotary motion; in a coal-fired plant, heat produced by burning coal is used to convert water to steam, and this steam is directed against the turbine blades. As the loop rotates, the magnetic flux through it changes with time, inducing an emf and a current in an external circuit. The ends of the loop are connected to slip rings that rotate with the loop. Connections to the external circuit are made by stationary brushes in contact with the slip rings.

We can derive an expression for the emf generated in the rotating loop by making use of the equation for motional emf, $\varepsilon = BlV$. Figure 6.9a shows a loop of wire rotating clockwise in a uniform magnetic field directed to the right. The magnetic force (qvB) on the charges in wires AB and CD is



Active Figure 6.18

not along the lengths of the wires. (The force on the electrons in these wires is perpendicular to the wires.) Hence, an emf is generated only in wires BC and AD. At any instant, wire BC has velocity $\vec{V}$ at an angle u with the magnetic field, as shown in Figure 6.19b. (Note that the component of velocity parallel to the field has no effect on the charges in the wire, whereas the component of velocity perpendicular to the field produces a magnetic force on the charges that moves electrons from C to B.)



Active Figure 6.19

The emf generated in wire BC equals $Blv_\perp$ where l is the length of the wire and $v_\perp$ is the component of velocity perpendicular to the field. An emf of $Blv_\perp$ is also generated in wire DA, and the sense of this emf is the same as that in wire BC. Because $v_\perp = v\sin\theta$, the total induced emf is

$$\mathcal{E} = 2B\ell v_\perp = 2B\ell v \sin\theta$$

[6.6]

If the loop rotates with a constant angular speed v, we can use the relation $\theta = vt$ in Equation 6.6. Furthermore, because every point on the wires BC and DA rotates in a circle about the axis of rotation with the same angular speed v, we have $v = r\omega = (a/2)\,\omega$, where a is the length of sides AB and CD. Equation 6.6 therefore reduces to

$$\mathcal{E} = 2B\ell\left(\frac{a}{2}\right)\omega \sin\omega t = B\ell a\omega \sin\omega t$$

If a coil has N turns, the emf is N times as large because each loop has the same emf induced in it. Further, because the area of the loop is A = la, the total emf is

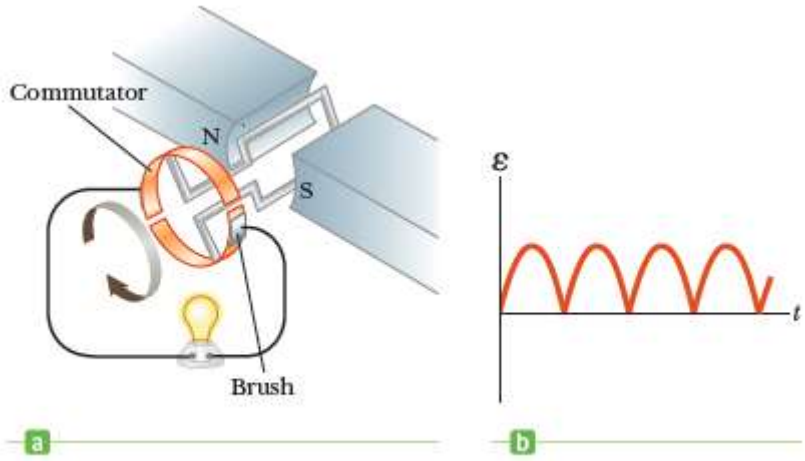
$$\mathcal{E} = NBA\omega \sin \omega t \qquad [6.7]$$

This result shows that the emf varies sinusoidally with time, as plotted in Active Figure 6.18b. Note that the maximum emf has the value

$$\mathcal{E}_{max} = NBA\omega \qquad [6.8]$$

which occurs when $\omega t = 90°$ or 270°. In other words, $\varepsilon = \varepsilon_{max}$ when the plane of the loop is parallel to the magnetic field. Further, the emf is zero when $\omega t = 0$ or 180°, which happens whenever the magnetic field is perpendicular to the plane of the loop. In the United States and Canada the frequency of rotation for commercial generators is 60 Hz, whereas in some European countries 50 Hz is used. (Recall that $\omega = 2\pi f$, where f is the frequency in hertz.)

The direct-current (DC) generator is illustrated in Active Figure 6.20a. The components are essentially the same as those of the AC generator except that the contacts to the rotating loop are made by a split ring, or commutator. In this design the output voltage always has the same polarity and the current is a pulsating direct current, as in Active Figure 6.20b. Note that the contacts to the split ring reverse their roles every half cycle. At the same time, the polarity of the induced emf reverses. Hence, the polarity of the split ring remains the same.



Active Figure 20.21

A pulsating DC current is not suitable for most applications. To produce a steady DC current, commercial DC generators use many loops and commutators distributed around the axis of rotation so that the sinusoidal pulses from the loops overlap in phase. When these pulses are superimposed, the DC output is almost free of fluctuations.

PROBLEM An AC generator consists of eight turns of wire, each having area A = 0.090 0 m^2, with a total resistance of 12.0 V. The coil rotates in a magnetic field of 0.500 T at a constant frequency of 60.0 Hz, with axis of rotation perpendicular to the direction of the magnetic field. (a) Find the maximum induced emf. (b) What is the maximum induced current? (c) Determine the induced emf and current as functions of time. (d) What maximum torque must be applied to keep the coil turning?

STRATEGY From the given frequency, calculate the angular frequency ω and substitute it, together with given quantities, into Equation 6.8. As functions of time, the emf and current have the form Asinωt, where A is the maximum emf or current, respectively. For part (d), calculate the magnetic torque on the coil when the current is at a maximum. The applied torque must do work against this magnetic torque to keep the coil turning.

SOLUTION

(a) Find the maximum induced emf.

First, calculate the angular frequency of the rotational motion:

$$\omega = 2\pi f = 2\pi(60.0\text{ Hz}) = 377\text{ rad/s}$$

Substitute the values for N, A, B, and ω into Equation 20.8, obtaining the maximum induced emf:

$$\mathcal{E}_{max} = NAB\omega = 8(0.090\,0\text{ m}^2)(0.500\text{ T})(377\text{ rad/s}) = 136\text{ V}$$

(b) What is the maximum induced current?

Substitute the maximum induced emf $\mathcal{E}_{max}$ and the resistance R into Ohm's law to find the maximum induced current:

$$I_{max} = \frac{\mathcal{E}_{max}}{R} = \frac{136\text{ V}}{12.0\,\Omega} = 11.3\text{ A}$$

(c) Determine the induced emf and the current as functions of time.

Substitute $\mathcal{E}_{max}$ and ω into Equation 20.7 to obtain the variation of $\mathcal{E}$ with time t in seconds:

$$\mathcal{E} = \mathcal{E}_{max}\sin\omega t = (136\text{ V})\sin 377t$$

The time variation of the current looks just like this expression, except with the maximum current out in front:

$$I = (11.3\text{ A})\sin 377t$$

(d) Calculate the maximum applied torque necessary to keep the coil turning.

Write the equation for magnetic torque:

$$\tau = \mu B\sin\theta$$

Calculate the maximum magnetic moment of the coil, μ:

$$\mu = I_{max}AN = (11.3\text{ A})(0.090\text{ m}^2)(8) = 8.14\text{ A}\cdot\text{m}^2$$

Substitute into the magnetic torque equation, with $\theta = 90°$ to find the maximum applied torque:

$$\tau_{max} = (8.14\text{ A}\cdot\text{m}^2)(0.500\text{ T})\sin 90° = 4.07\text{ N}\cdot\text{m}$$

Motors and Back emf

Motors are devices that convert electrical energy to mechanical energy. Essentially, a motor is a generator run in reverse: instead of a current being generated by a rotating loop, a current is supplied to the loop by a source of emf, and the magnetic torque on the current-carrying loop causes it to rotate. A motor can perform useful mechanical work when a shaft connected to its rotating coil is attached to some external device. As the coil in the motor rotates, however, the changing magnetic flux through it induces an emf that acts to reduce the current in the coil. If it increased the current, Lenz's law would be violated. The phrase back emf is used for an emf that tends to reduce the applied current. The back emf increases in magnitude as the rotational speed of the coil increases. We can picture this state of affairs as the equivalent circuit in Figure 6.21. For illustrative purposes, assume the external power source supplying current in the coil of the motor has a voltage of 120 V, the coil has a resistance of 10 V, and the back emf induced in the coil at this instant is 70 V. The voltage available to supply current equals the difference between the applied voltage and the back emf, or 50 V in this case. The current is always reduced by the back emf. When a motor is turned on, there is no back emf initially and the current is very large because it's limited only by the resistance of the coil. As the coil begins to rotate, the induced back emf opposes the applied voltage and the current in the coil is reduced. If the mechanical load increases, the motor slows down, which decreases the back emf. This reduction in the back emf increases the current in the coil and therefore also increases the power needed from the external voltage source. As a result, the power requirements

for starting a motor and for running it under heavy loads are greater than those for running the motor under average loads. If the motor is allowed to run under no mechanical load, the back emf reduces the current to a value just large enough to balance energy losses by heat and friction.

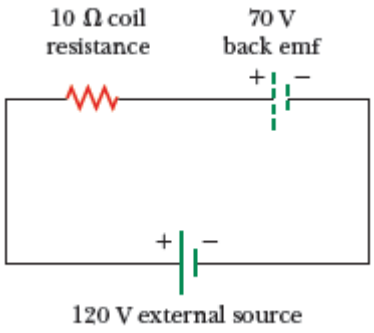


Figure 6.21

6.5 Self-Inductance

Consider a circuit consisting of a switch, a resistor, and a source of emf, as in Figure 6.22. When the switch is closed, the current doesn't immediately change from zero to its maximum value, ε/R. The law of electromagnetic induction, Faraday's law, prevents this change. What happens instead is the following: as the current increases with time, the magnetic flux through the loop due to this current also increases. The increasing flux induces an emf in the circuit that opposes the change in magnetic flux. By Lenz's law, the induced emf is in the direction indicated by the dashed battery in the figure. The net potential difference across the resistor is the emf of the battery minus the opposing induced emf. As the magnitude of the current increases, the rate of increase lessens and hence the induced emf decreases. This opposing emf results in a gradual increase in the current. For the same reason, when the switch is opened, the current doesn't immediately fall to zero. This effect is called self-induction because the changing flux through the circuit arises from the circuit itself. The emf that is set up in the circuit is called a **self-induced emf**.

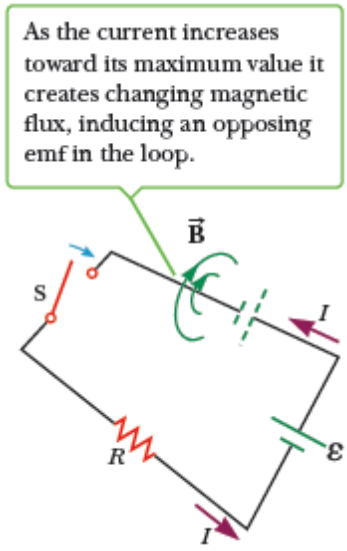


Figure 6.22

As a second example of self-inductance, consider Figure 6.23, which shows a coil wound on a cylindrical iron core. (A practical device would have several hundred turns.) Assume the current changes with time. When the current is in the direction shown, a magnetic field is set up inside the coil, directed from right to left. As a result, some lines of magnetic flux pass through the cross-sectional area of the coil. As the current changes with time, the flux through the coil changes and induces an emf in the coil. Lenz's law shows that this induced emf has a direction that opposes the change in the current. If the current is increasing, the induced emf is as pictured in Figure 20.24b, and if the current is decreasing, the induced emf is as shown in Figure 20.24c.

To evaluate self-inductance quantitatively, first note that, according to Faraday's law, the induced emf is given by Equation 6.2:

$$\mathcal{E} = -N\frac{\Delta\Phi_B}{\Delta t}$$

The magnetic flux is proportional to the magnetic field, which is proportional to the current in the coil. Therefore**, the self-induced emf must be proportional to the rate of change of the current with time**, or

$$\mathcal{E} = -L\frac{\Delta I}{\Delta t} \qquad [6.9]$$

where L is a proportionality constant called the **inductance** of the device. The negative sign indicates that a changing current induces an emf in opposition to the change. In other words, if the current is increasing (ΔI positive), the induced emf is negative, indicating opposition to the increase in current. Likewise, if the current is decreasing (ΔI negative), the sign of the induced emf is positive, indicating that the emf is acting to oppose the decrease. The inductance of a coil depends on the cross-sectional area of the coil and other quantities, which can all be grouped under the general heading of geometric factors. The SI unit of inductance is the **henry** (H), which, from Equation 6.9, is equal to 1 volt-second per ampere:

1 H = 1 V . s/A

In the process of calculating self-inductance, it is often convenient to equate Equations 6.2 and 6.9 to find an expression for L:

$$N\frac{\Delta\Phi_B}{\Delta t} = L\frac{\Delta I}{\Delta t} \qquad L = N\frac{\Delta\Phi_B}{\Delta I} = \frac{N\Phi_B}{I} \qquad [6.10]$$

In general, determining the inductance of a given current element can be challenging. Finding an expression for the inductance of a common solenoid, however, is straightforward. Let the solenoid have N turns and length l. Assume l is large compared with the radius and the core of the solenoid is air. We take the interior magnetic field to be uniform and given by,

$$B = \mu_0 nI = \mu_0\frac{N}{\ell}I$$

where $n = N/l$ is the number of turns per unit length. The magnetic flux through each turn is therefore

$$\Phi_B = BA = \mu_0\frac{N}{\ell}AI$$

where A is the cross-sectional area of the solenoid. From this expression and Equation 6.10, we find

$$L = \frac{N\Phi_B}{I} = \frac{\mu_0 N^2 A}{\ell} \qquad [6.11a]$$

This equation shows that L depends on the geometric factors l and A and on μ_0 and is proportional to the square of the number of turns. Because N = nl, we can also express the result in the form

$$L = \mu_0\frac{(n\ell)^2}{\ell}A = \mu_0 n^2 A\ell = \mu_0 n^2 V \qquad [6.11b]$$

where V = Al is the volume of the solenoid.

PROBLEM (a) Calculate the inductance of a solenoid containing 300 turns if the length of the solenoid is 25.0 cm and its cross-sectional area is 4.00 × 10^{-4} m^2. (b) Calculate the self-induced emf in the solenoid described in part (a) if the current in the solenoid decreases at the rate of 50.0 A/s.

STRATEGY Substituting given quantities into Equation 6.11a gives the inductance L. For part (b), substitute the result of part (a) and ΔI/Δt = -50.0 A/s into Equation 6.9 to get the self-induced emf.

SOLUTION

(a) Calculate the inductance of the solenoid.

Substitute the number N of turns, the area A, and the length ℓ into Equation 20.11a to find the inductance:

$$L = \frac{\mu_0 N^2 A}{\ell}$$

$$= (4\pi \times 10^{-7}\ \mathrm{T \cdot m/A})\frac{(300)^2(4.00 \times 10^{-4}\ \mathrm{m^2})}{25.0 \times 10^{-2}\ \mathrm{m}}$$

$$= 1.81 \times 10^{-4}\ \mathrm{T \cdot m^2/A} = 0.181\ \mathrm{mH}$$

(b) Calculate the self-induced emf in the solenoid.

Substitute L and $\Delta I/\Delta t = -50.0\ \mathrm{A/s}$ into Equation 20.9, finding the self-induced emf:

$$\mathcal{E} = -L\frac{\Delta I}{\Delta t} = -(1.81 \times 10^{-4}\ \mathrm{H})(-50.0\ \mathrm{A/s})$$

$$= 9.05\ \mathrm{mV}$$

REMARKS Notice that ΔI/Δt is negative because the current is decreasing with time. The expression for the inductance in part (a) relies on the assumption that the radius of the solenoid is small compared to its length.

6.6 RL Circuits

A circuit element that has a large inductance, such as a closely wrapped coil of many turns, is called an **inductor**. The circuit symbol for an inductor is —ooo—. We will always assume the self-inductance of the remainder of the circuit is negligible compared with that of the inductor in the circuit. To gain some insight into the effect of an inductor in a circuit, consider the two circuits in Figure 6.23. Figure 6.23a shows a resistor connected to the terminals of a battery. For this circuit, Kirchhoff's loop rule is $\varepsilon - IR = 0$. The voltage drop across the resistor is

$$\Delta V_R = -IR \qquad [6.12]$$

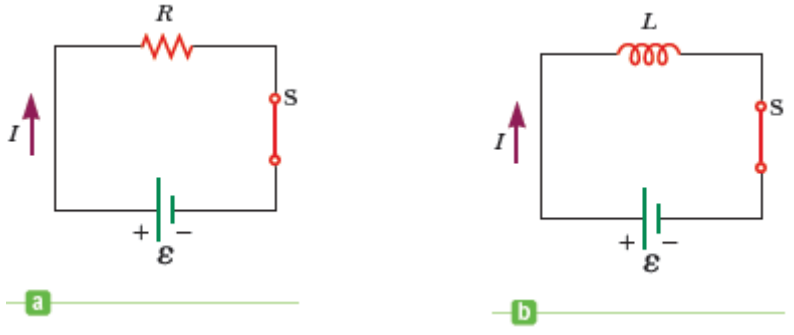
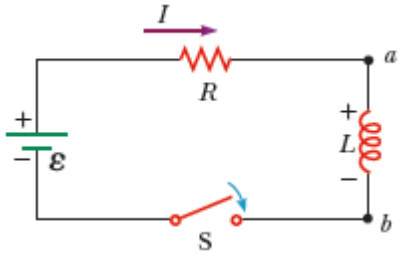


Figure 6.23

In this case, we interpret resistance as a measure of opposition to the current. Now consider the circuit in Figure 6.23b, consisting of an inductor connected to the terminals of a battery. At the instant the switch in this circuit is closed, because $IR = 0$, the emf of the battery equals the back emf generated in the coil. Hence, we have

$$\mathcal{E}_L = -L\frac{\Delta I}{\Delta t} \qquad [6.13]$$

From this expression, **we can interpret L as a measure of opposition to the rate of change of current**. Active Figure 6.24 shows a circuit consisting of a resistor, an inductor, and a battery. Suppose the switch is closed at t = 0.



Active Figure 6.24

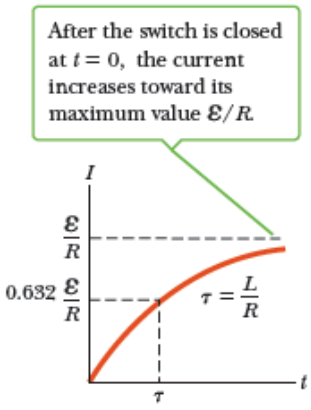
The current begins to increase, but the inductor produces an emf that opposes the increasing current. As a result, the current can't change from zero to its maximum value of ε/R instantaneously. Equation 6.13 shows that the induced emf is a maximum when the current is changing most rapidly, which occurs when the switch is first closed. As the current approaches its steady-state value, the back emf of the coil falls off because the current is changing more slowly. Finally, when the current reaches its steady-state value, the rate of change is zero and the back emf is also zero. Active Figure 6.25 plots current in the circuit as a function of time. This plot is similar to that of the charge on a capacitor as a function of time in connection with RC circuits. In that case, we found it convenient to introduce a quantity called the time constant of the circuit, which told us something about the time required for the capacitor to approach its steady-state charge. In the same way, time constants are defined for circuits containing resistors and inductors. The **time constant** t for an RL circuit is the time required for the current in the circuit to reach 63.2% of its final value ε/R; the time constant of an RL circuit is given by

$$\tau = \frac{L}{R} \qquad [6.14]$$

Using methods of calculus, it can be shown that the current in such a circuit is given by

$$I = \frac{\mathcal{E}}{R}(1 - e^{-t/\tau}) \qquad [6.15]$$

This equation is consistent with our intuition: when the switch is closed at t = 0, the current is initially zero, rising with time to some maximum value, which features



Active Figure 6.25

PROBLEM A 12.6-V battery is in a circuit with a 30.0-mH inductor and a 0.150-V resistor, as in Active Figure 6.24. The switch is closed at t 5 0. (a) Find the time constant of the circuit. (b) Find the current after one time constant has elapsed.(c) Find the voltage drops across the resistor when t = 0 and t = one time constant. (d) What's the rate of change of the current after one time constant?

STRATEGY Part (a) requires only substitution into the definition of time constant. With this value and Ohm's law, the current after one time constant can be found, and multiplying this current by the resistance yields the voltage drop across the resistor after one time constant. With the voltage drop and Kirchhoff's loop law, the voltage across the inductor can be found. This value can be substituted into Equation 6.13 to obtain the rate of change of the current.

SOLUTION

(a) What's the time constant of the circuit?

Substitute the inductance L and resistance R into Equation 20.14, finding the time constant:

$$\tau = \frac{L}{R} = \frac{30.0 \times 10^{-3}\,\text{H}}{0.150\ \Omega} = 0.200\ \text{s}$$

(b) Find the current after one time constant has elapsed.

First, use Ohm's law to compute the final value of the current after many time constants have elapsed:

$$I_{\max} = \frac{\mathcal{E}}{R} = \frac{12.6\ \text{V}}{0.150\ \Omega} = 84.0\ \text{A}$$

After one time constant, the current rises to 63.2% of its final value:

$$I_{1\tau} = (0.632)I_{\max} = (0.632)(84.0\ \text{A}) = 53.1\ \text{A}$$

(c) Find the voltage drops across the resistance when $t = 0$ and $t =$ one time constant.

Initially, the current in the circuit is zero, so, from Ohm's law, the voltage across the resistor is zero:

$$\Delta V_R = IR$$
$$\Delta V_R\,(t = 0\ \text{s}) = (0\ \text{A})(0.150\ \Omega) = 0$$

Next, using Ohm's law, find the magnitude of the voltage drop across the resistor after one time constant:

$$\Delta V_R\,(t = 0.200\ \text{s}) = (53.1\ \text{A})(0.150\ \Omega) = 7.97\ \text{V}$$

(d) What's the rate of change of the current after one time constant?

Using Kirchhoff's voltage rule, calculate the voltage drop across the inductor at that time:

$$\mathcal{E} + \Delta V_R + \Delta V_L = 0$$

Solve for ΔV_L:

$$\Delta V_L = -\mathcal{E} - \Delta V_R = -12.6\ \text{V} - (-7.97\ \text{V}) = -4.6\ \text{V}$$

Now solve Equation 20.13 for $\Delta I/\Delta t$ and substitute:

$$\Delta V_L = -L\frac{\Delta I}{\Delta t}$$
$$\frac{\Delta I}{\Delta t} = -\frac{\Delta V_L}{L} = -\frac{-4.6\ \text{V}}{30.0 \times 10^{-3}\,\text{H}} = 150\ \text{A/s}$$

6.7 Energy Stored in a Magnetic Field

The emf induced by an inductor prevents a battery from establishing an instantaneous current in a circuit. The battery has to do work to produce a current. We can think of this needed work as energy stored in the inductor in its magnetic field. In a manner similar to that used to find the energy stored in a capacitor, we find that the energy stored by an inductor is

$$PE_L = \tfrac{1}{2}LI^2 \qquad [6.16]$$

PROBLEM A 12.0-V battery is connected in series to a 25.0-V resistor and a 5.00-H inductor. (a) Find the maximum current in the circuit. (b) Find the energy stored in the inductor at this time. (c) How much energy is stored in the inductor when the current is changing at a rate of 1.50 A/s?

STRATEGY In part (a) Ohm's law and Kirchhoff's voltage rule yield the maximum current because the voltage across the inductor is zero when the current is maximal. Substituting the current into Equation 6.16 gives the energy stored in the inductor. In part (c) the given rate of change of the current can be used to calculate the voltage drop across the inductor at the specified time. Kirchhoff's voltage rule and Ohm's law then give the current I at that time, which can be used to find the energy stored in the inductor.

SOLUTION

(a) Find the maximum current in the circuit.

Apply Kirchhoff's voltage rule to the circuit:

$$\Delta V_{\text{batt}} + \Delta V_R + \Delta V_L = 0$$

$$\mathcal{E} - IR - L\frac{\Delta I}{\Delta t} = 0$$

When the maximum current is reached, $\Delta I/\Delta t$ is zero, so the voltage drop across the inductor is zero. Solve for the maximum current I_{max}:

$$I_{\text{max}} = \frac{\mathcal{E}}{R} = \frac{12.0\ \text{V}}{25.0\ \Omega} = 0.480\ \text{A}$$

(b) Find the energy stored in the inductor at this time.

Substitute known values into Equation 20.16:

$$PE_L = \tfrac{1}{2}LI_{\text{max}}^2 = \tfrac{1}{2}(5.00\ \text{H})(0.480\ \text{A})^2 = 0.576\ \text{J}$$

(c) Find the energy in the inductor when the current changes at a rate of 1.50 A/s.

Apply Kirchhoff's voltage rule to the circuit once again:

$$\mathcal{E} - IR - L\frac{\Delta I}{\Delta t} = 0$$

Solve this equation for the current I and substitute:

$$I = \frac{1}{R}\left(\mathcal{E} - L\frac{\Delta I}{\Delta t}\right)$$

$$= \frac{1}{25.0\ \Omega}[12.0\ \text{V} - (5.00\ \text{H})(1.50\ \text{A/s})] = 0.180\ \text{A}$$

Finally, substitute the value for the current into Equation 20.15, finding the energy stored in the inductor:

$$PE_L = \tfrac{1}{2}LI^2 = \tfrac{1}{2}(5.00\ \text{H})(0.180\ \text{A})^2 = 0.081\ 0\ \text{J}$$

PROBLEMS

1. A uniform magnetic field of magnitude 0.50 T is directed perpendicular to the plane of a rectangular loop having dimensions 8.0 cm by 12 cm. Find the magnetic flux through the loop.

2. Find the flux of Earth's magnetic field of magnitude 5.00×10^{-5} T through a square loop of area 20.0 cm^2, (a) when the field is perpendicular to the plane of the loop, (b) when the field makes a 30.0° angle with the normal to the plane of the loop, and (c) when the field makes a 90.0° angle with the normal to the plane.

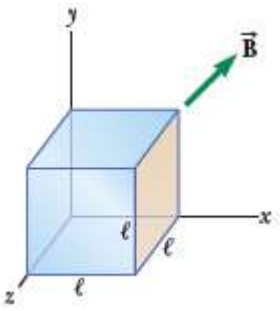
3. A circular loop of radius 12.0 cm is placed in a uniform magnetic field. (a) If the field is directed perpendicular to the plane of the loop and the magnetic flux through the loop is 8.00×10^{-3} T . m^2, what is the strength of the magnetic field? (b) If the magnetic field is directed parallel to the plane of the loop, what is the magnetic flux through the loop?

4. A long, straight wire carrying a current of 2.00 A is placed along the axis of a cylinder of radius 0.500 m and a length of 3.00 m. Determine the total magnetic flux through the cylinder.

5. A long, straight wire lies in the plane of a circular coil with a radius of 0.010 m. The wire carries a current of 2.0 A and is placed along a diameter of the coil. (a) What is the net flux through the coil? (b) If the wire passes through the center of the coil and is perpendicular to the plane of the coil, what is the net flux through the coil?

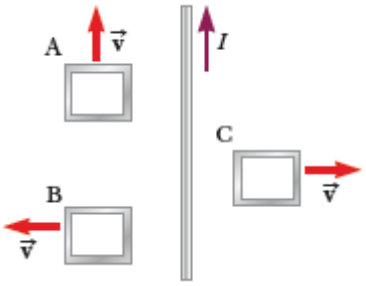
6. A 400-turn solenoid of length 36.0 cm and radius 3.00 cm carries a current of 5.00 A. Find (a) the magnetic field strength inside the coil at its midpoint and (b) the magnetic flux through a circular cross-sectional area of the solenoid at its midpoint.

7. A cube of edge length l = 2.5 cm is positioned as shown. There is a uniform magnetic field throughout the region with components B_x = +5.0 T, B_y = +4.0 T, and B_z = +3.0 T. (a) Calculate the flux through the shaded face of the cube. (b) What is the total flux emerging from the volume enclosed by the cube (i.e., the total flux through all six faces)?

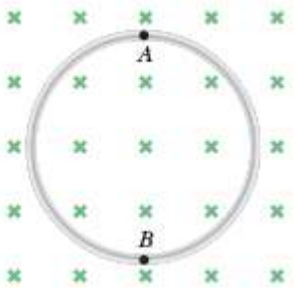


8. Transcranial magnetic stimulation (TMS) is a noninvasive technique used to stimulate regions of the human brain. A small coil is placed on the scalp, and a brief burst of current in the coil produces a rapidly changing magnetic field inside the brain. The induced emf can be sufficient to stimulate neuronal activity. One such device generates a magnetic field within the brain that rises from zero to 1.5 T in 120 ms. Determine the induced emf within a circle of tissue of radius 1.6 mm and that is perpendicular to the direction of the field.

9. Three loops of wire move near a long straight wire carrying a current as in the following Figure. What is the direction of the induced current, if any, in (a) loop A, (b) loop B, and (c) loop C.



10. The flexible loop in following Figure has a radius of 12 cm and is in a magnetic field of strength 0.15 T. The loop is grasped at points A and B and stretched until its area is nearly zero. If it takes 0.20 s to close the loop, what is the magnitude of the average induced emf in it during this time?



11. A wire loop of radius 0.30 m lies so that an external magnetic field of magnitude 0.30 T is perpendicular to the loop. The field reverses its direction, and its magnitude changes to 0.20 T in 1.5 s. Find the magnitude of the average induced emf in the loop during this time.

12. A circular loop of wire of radius 12.0 cm is placed in a magnetic field directed perpendicular to the plane of the loop, as shown in Figure problem 10. If the field decreases at the rate of 0.050 0 T/s in some time interval, what is the magnitude of the emf induced in the loop during this interval?

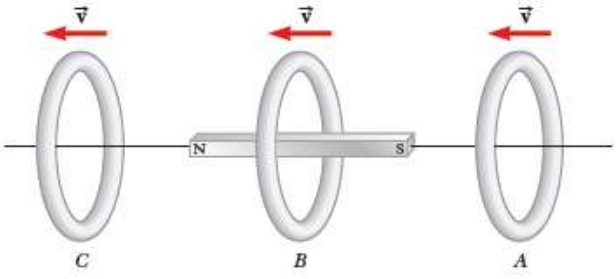
13. A technician wearing a circular metal band on his wrist moves his hand into a uniform magnetic field of magnitude 2.5 T in a time of 0.18 s. If the diameter of the band is 6.5 cm and the field is at an angle of 45° with the plane of the metal band while the hand is in the field, find the magnitude of the average emf induced in the band.

14. A truck is carrying a steel beam of length 15.0 m on a freeway. An accident causes the beam to be dumped off the truck and slide horizontally along the ground at a speed of 25.0 m/s. The velocity of the center of mass of the beam is northward while the length of the beam maintains an east–west orientation. The vertical component of the Earth's magnetic field at this location has a magnitude of 35.0 mT. What is the magnitude of the induced emf between the ends of the beam?

15. A 2.00-m length of wire is held in an east–west direction and moves horizontally to the north with a speed of 15.0 m/s. The vertical component of Earth's magnetic field in this region is 40.0 mT directed downward. Calculate the induced emf between the ends of the wire and determine which end is positive.

16. A pickup truck has a width of 79.8 in. If it is traveling north at 37 m/s through a magnetic field with vertical component of 35 mT, what magnitude emf is induced between the driver and passenger sides of the truck?

17. A bar magnet is held stationary while a circular loop of wire is moved toward the magnet at constant velocity at position A as in following Figure. The loop passes over the magnet at position B and moves away from the magnet at position C. Find the direction of the induced current in the loop using Lenz's law (a) at position A and (b) at position C. (c) What is the induced current in the loop at position B? Explain.



18. An automobile has a vertical radio antenna 1.20 m long. The automobile travels at 65.0 km/h on a horizontal road where Earth's magnetic field is 50.0 mT, directed toward the north and downward at an angle of 65.0° below the horizontal. (a) Specify the direction the automobile should move so as to generate the maximum motional emf in the antenna, with the top of the antenna positive relative to the bottom. (b) Calculate the magnitude of this induced emf.